



2026 SUMMER SCHOOL

Transforming Technology, Modern Practice, and Clinical Impact

JUNE 16–20 | UNIVERSITY OF MICHIGAN



Synthetic Images for ART

June 19th, 8:30-9:00am

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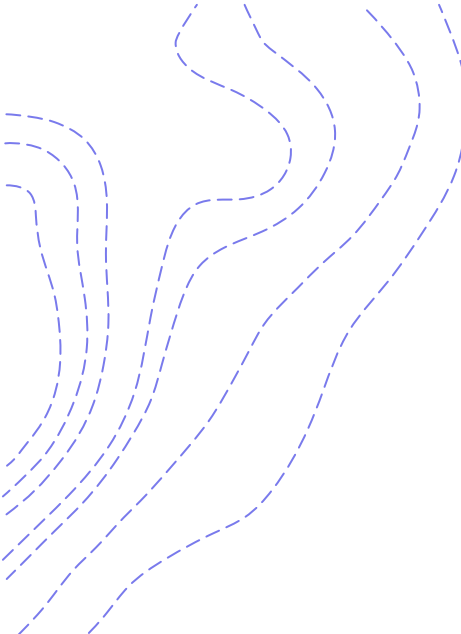
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Disclosures

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Siemens Healthineers and Varian Medical Systems





Introduction

Techniques

ART Applications

Validation & QA



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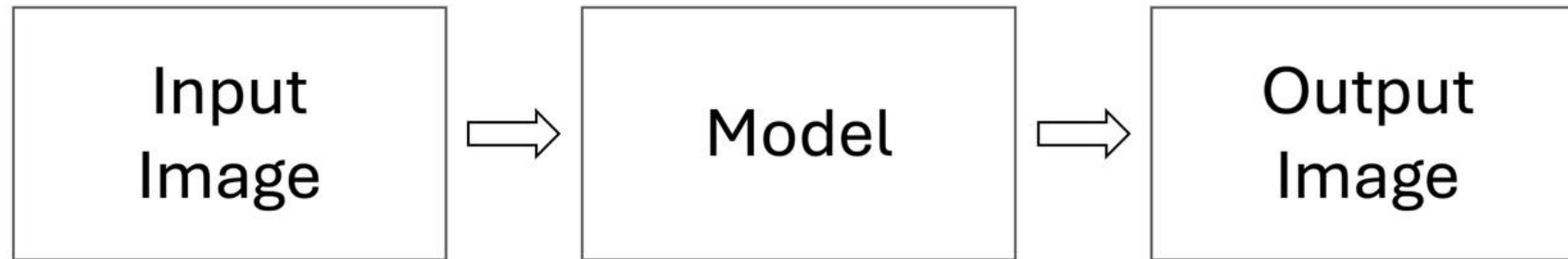


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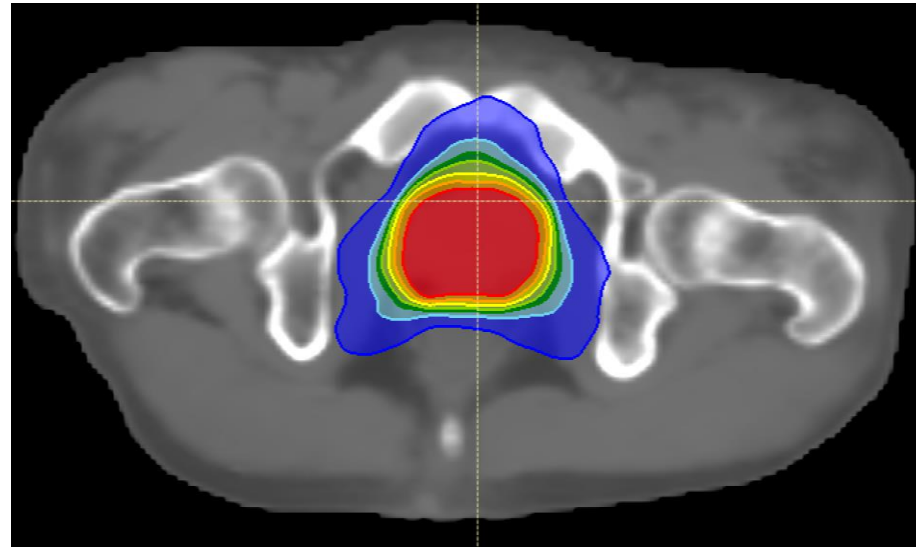
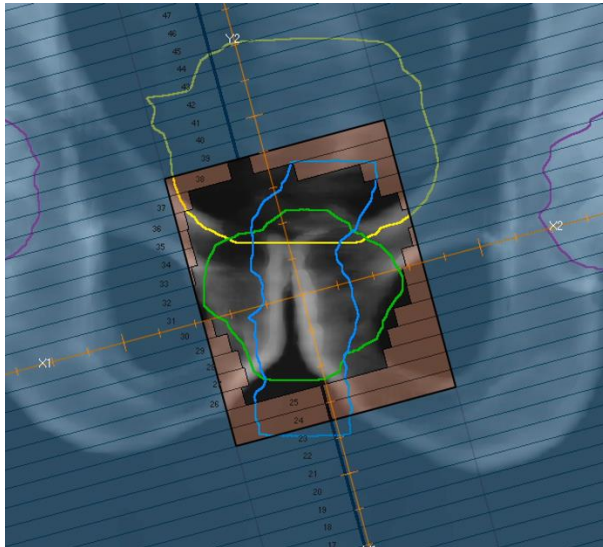
Synthetic Images



- Synthetic imaging techniques enable the **generation** of an output image from an input image using a computational model.
 - Input image is acquired on an imaging system.
 - Output image is assigned characteristics for a particular application.

Synthetic CT

- Most common application in radiotherapy is synthetic CT (sCT)
- CT dataset synthesized from another imaging dataset.
- sCTs can replace CTs in the radiotherapy workflow.
- MR-only planning, adaptive radiotherapy (ART), low-dose CT, PET/MRI, etc.



Techniques

Most common methods for synthetic image generation

1. Bulk density / Hounsfield Unit (HU) override
2. Deformable image registration (DIR) / Atlas-based
3. Deep learning



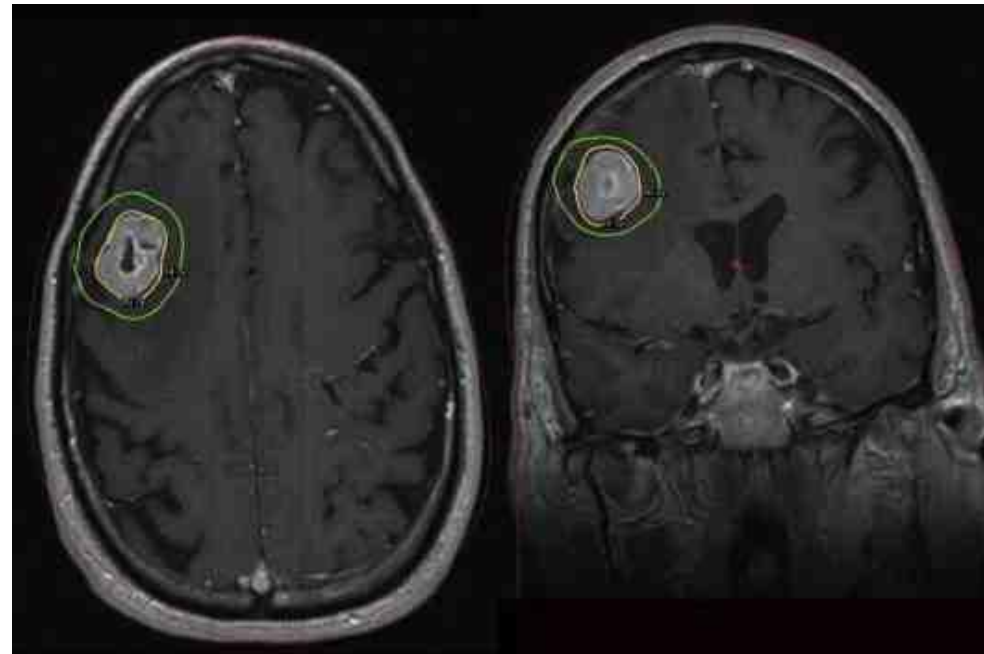
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1. Bulk Density / HU Override

- Pixel data of input image is overridden to CT HU values.
- Simplest approach: single HU value (e.g. GammaKnife)

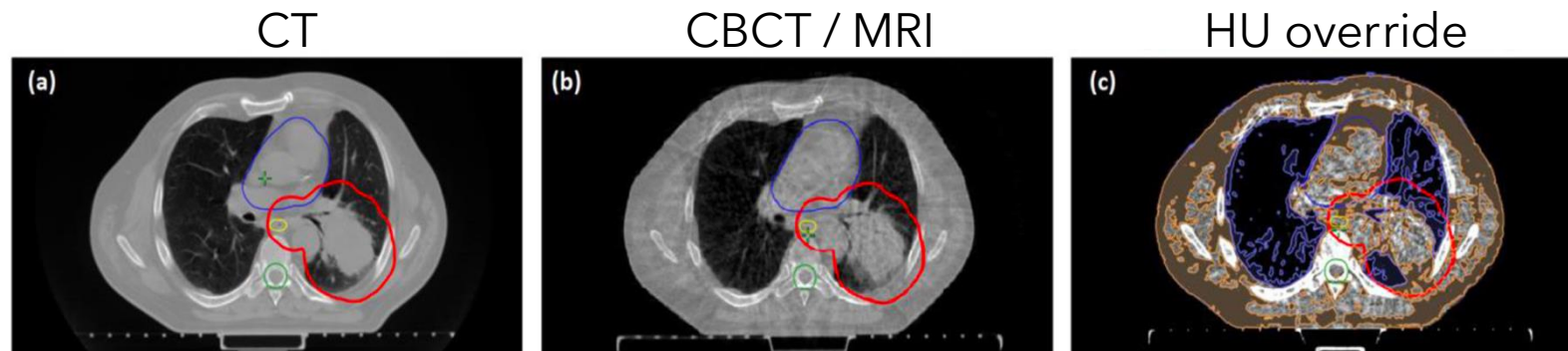


Lippitz et al. 2014

1. Bulk Density / HU Override

Multiple HU values assigned to different tissue classes (e.g. soft tissue, air, bone, etc.)

*Giacometti et al.
2019*

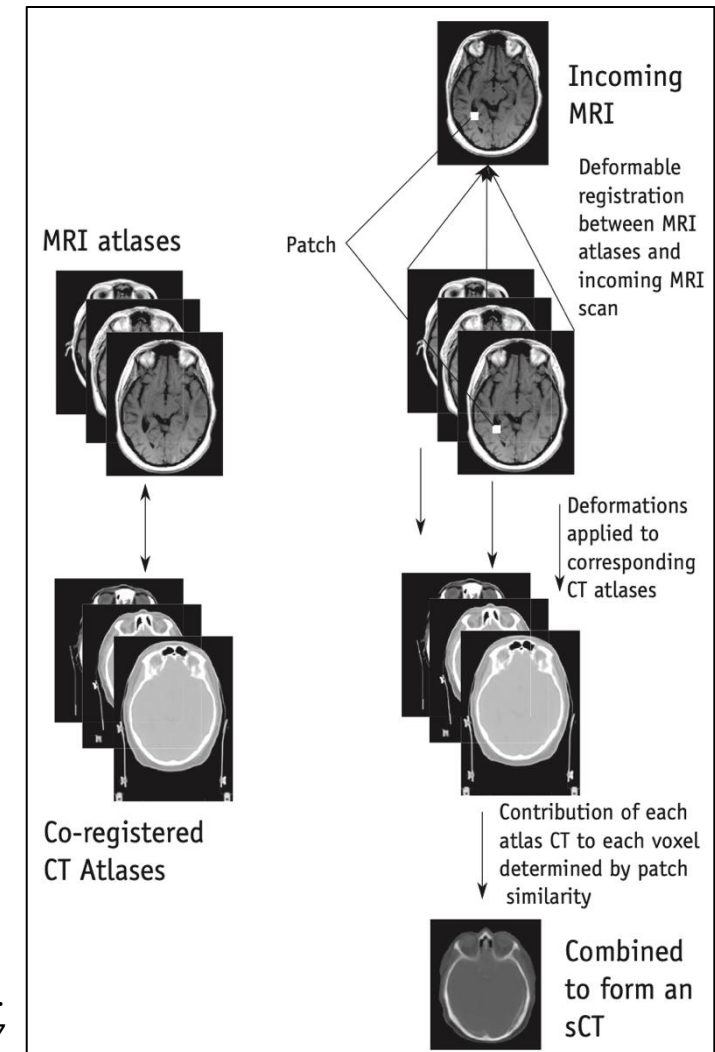


Scholey et al.



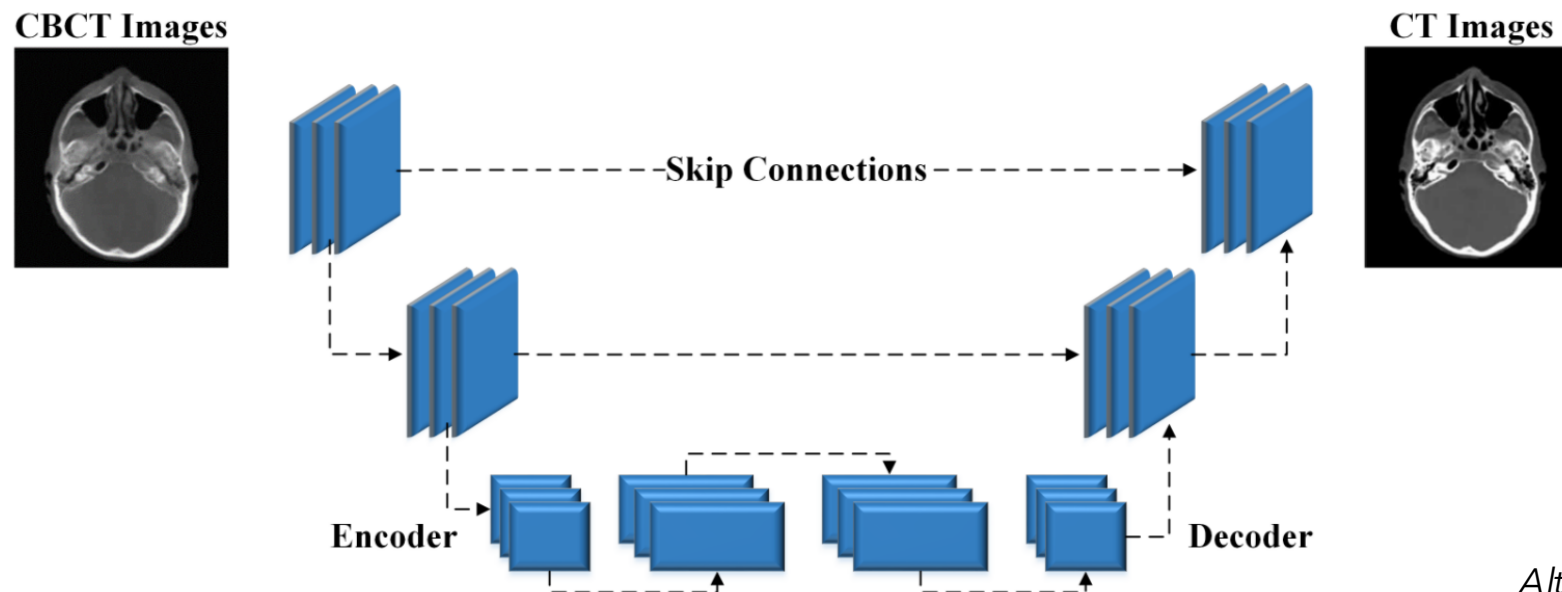
2. Deformable Image Registration

- DIR used for image synthesis by mapping anatomical information from one image onto another through voxel-by-voxel deformation fields.
- Also known as atlas-based approaches:
 - Simplest approaches utilize single or averaged atlas.
 - More advanced strategies use multiple atlases for DIR using patch-based or pattern recognition-based DIR approaches



3. Deep Learning

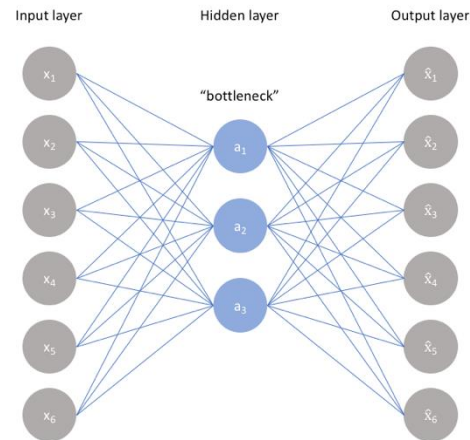
- Utilizes **neural networks** with many layers to extract image features.
- Typically consists of:
 - Training stage for the network to learn the mapping between the input and its target.
 - Inference stage to synthesize the target image from an input image.



3. Deep Learning

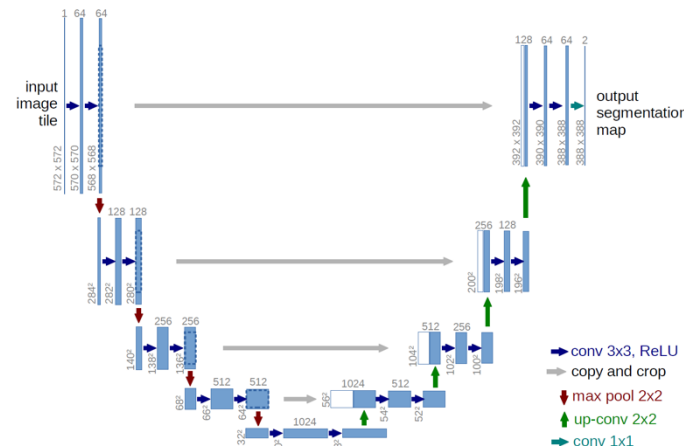
- Auto-encoder (AE), U-net, and generative adversarial network (GAN).
 - 90% articles are U-net or GAN studies.
 - GANs typically include AEs and/or U-Nets.

AE



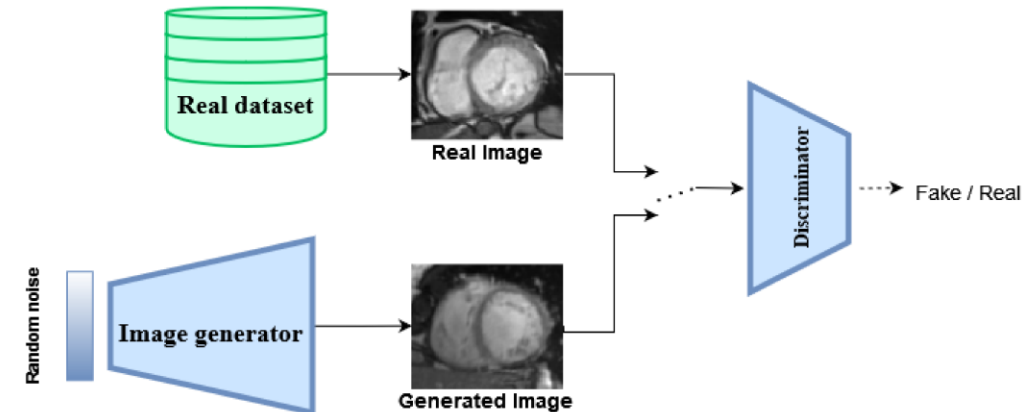
Jordan et al. 2018

U-Net



Ronneberger et al. 2015

GAN



Skandarani et al. 2023

Clinical ART Applications

- All three techniques have been used clinically:
 1. Bulk density / HU override
 2. DIR / Atlas-based
 3. Deep learning
- Many models use two or more of these algorithms.
- Two most common clinical applications:
 - MRI-based sCT
 - Conebeam CT (CBCT)-based sCT



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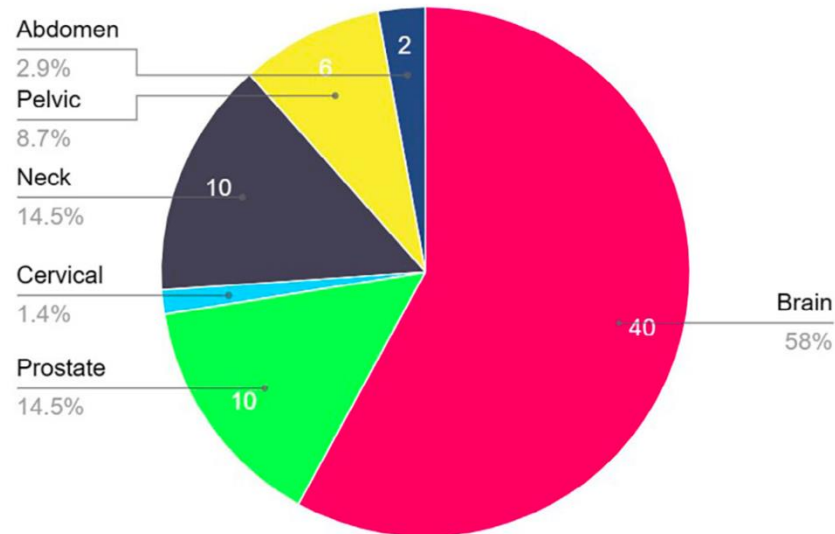


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MR-based sCT

- MRI-only radiotherapy
 - MRI → superior soft tissue contrast (contouring)
 - sCT → electron density (dose calculation)

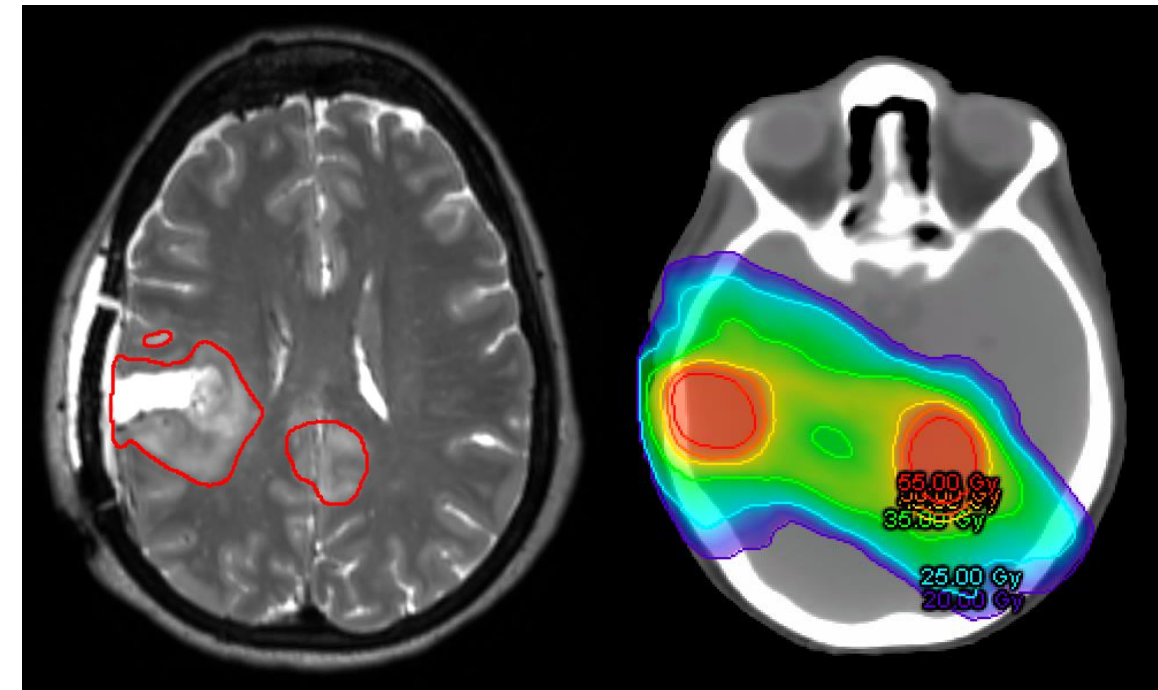
- Most common in brain and pelvis



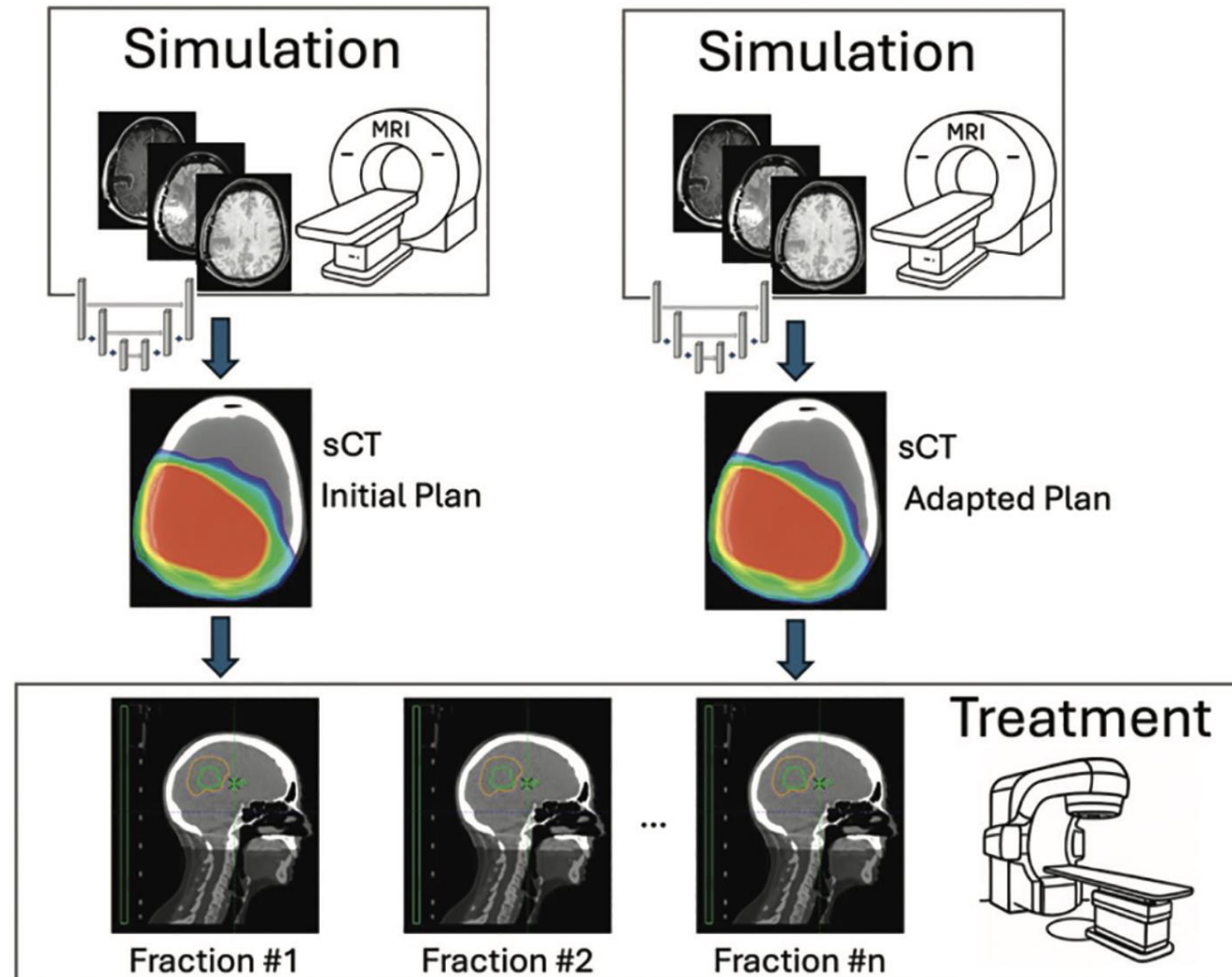
Bahloul et al. 2024

MRI

sCT



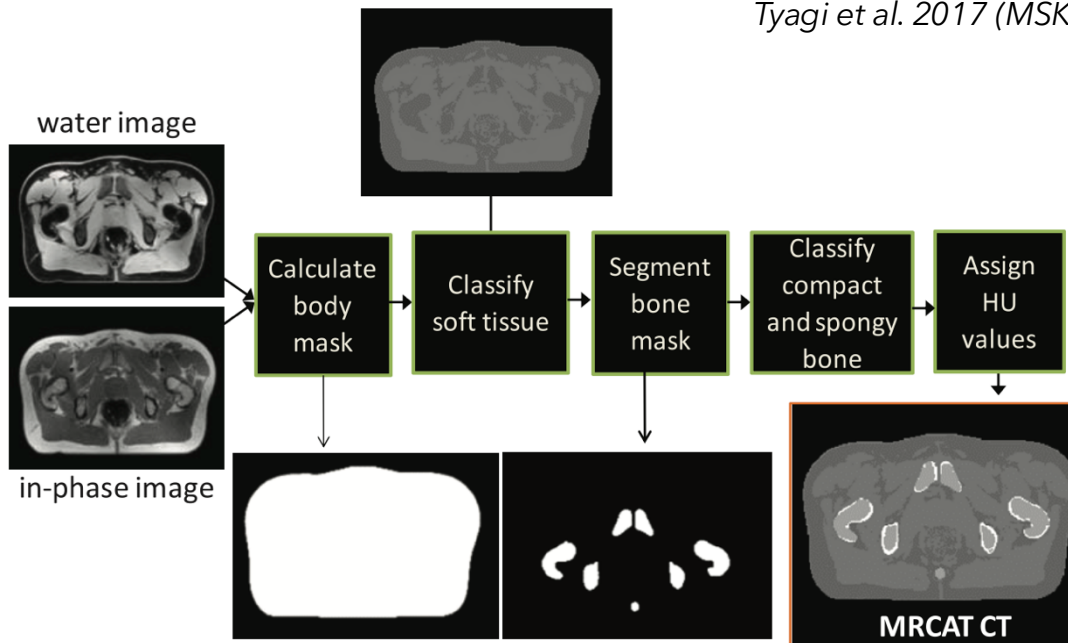
MR-based sCT (Offline Adaptive)



MR-only Radiation Therapy

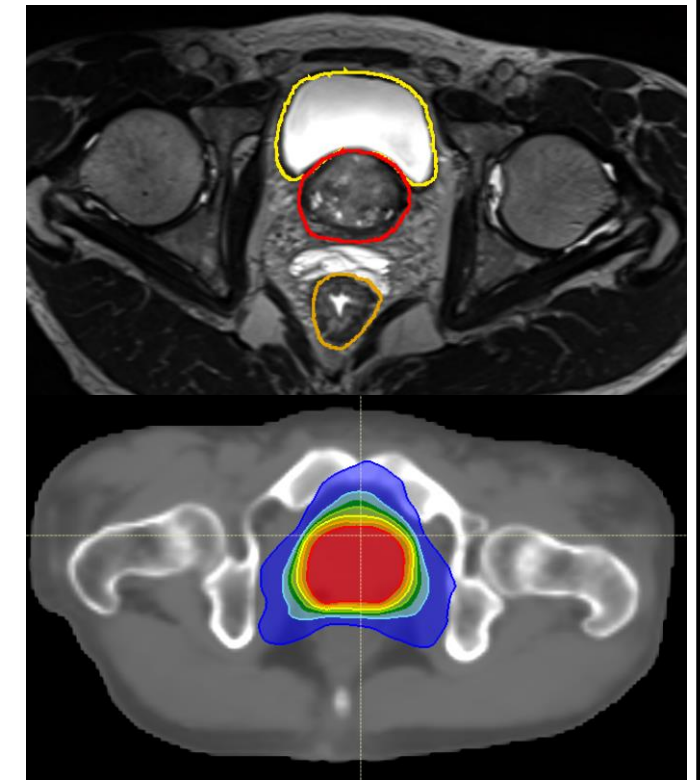
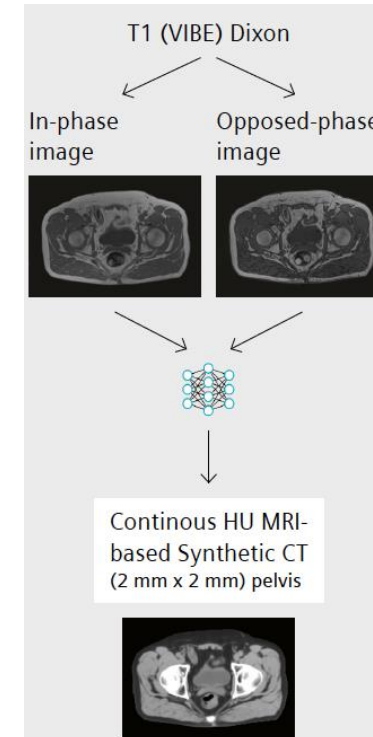
- First commercial application: **MRCAT** (MR for Calculating ATtenuation) (e.g. Phillips)
- Bulk HU approach (air, adipose, water, trabecular/spongy bone, compact/cortical bone)

Tyagi et al. 2017 (MSKCC)



Deep learning-based approach (e.g. Siemens)

Scholey et al. 2024 (UCSF)

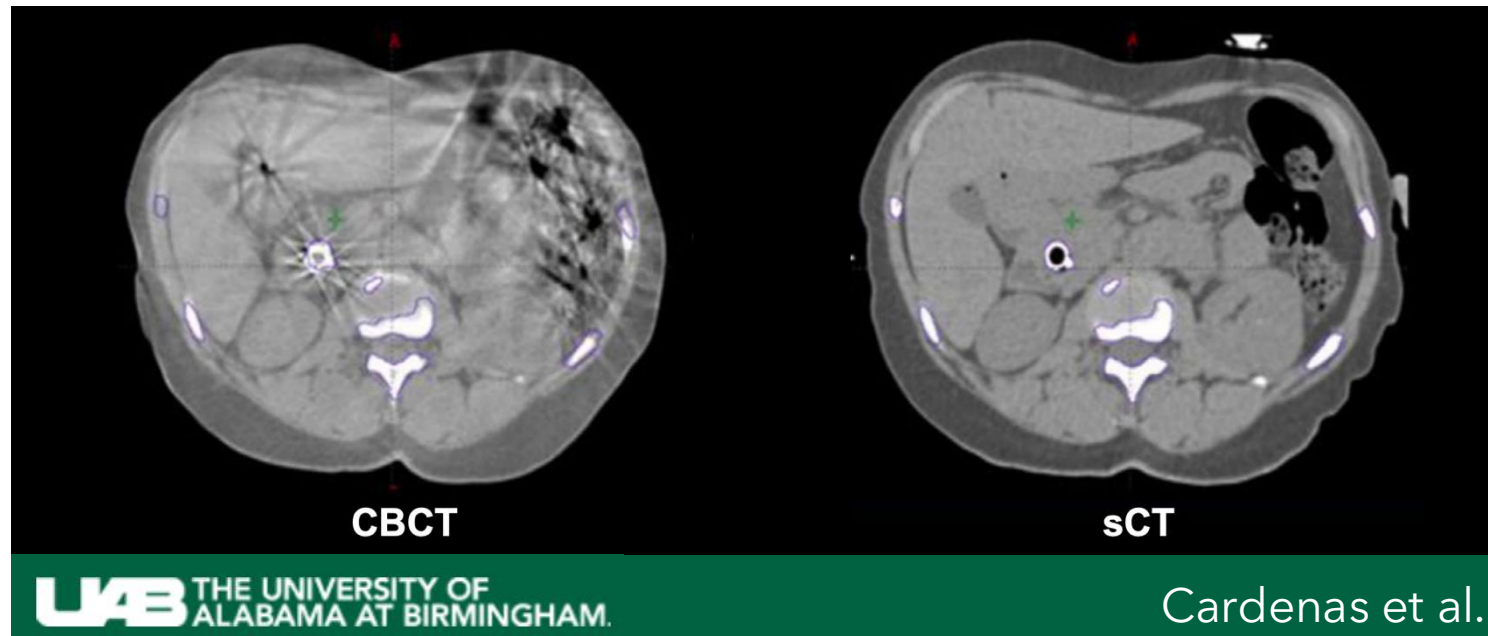
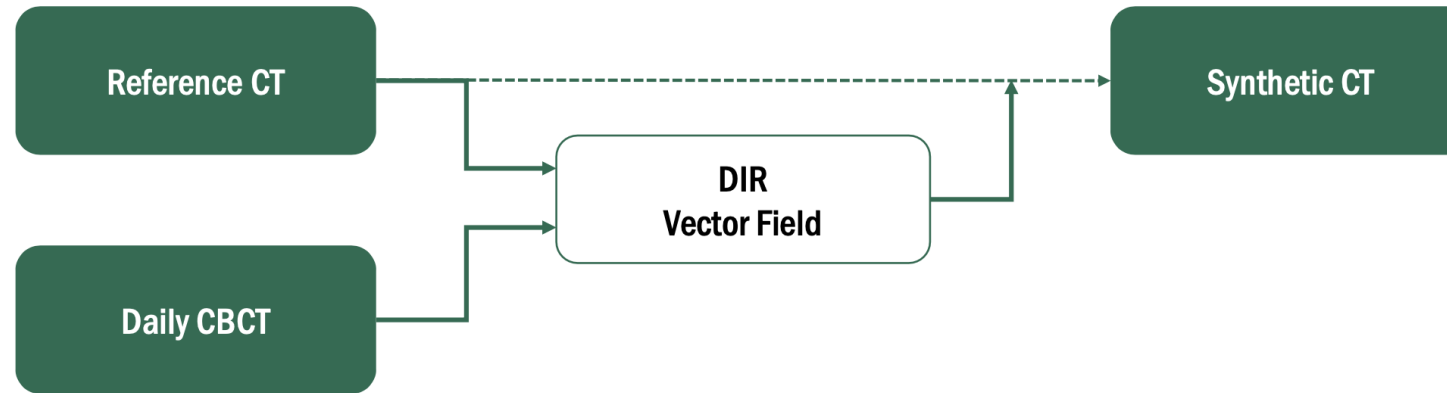


Online Adaptive RT

Synthetic image generated from reference and session images for adaptive dose calculation:

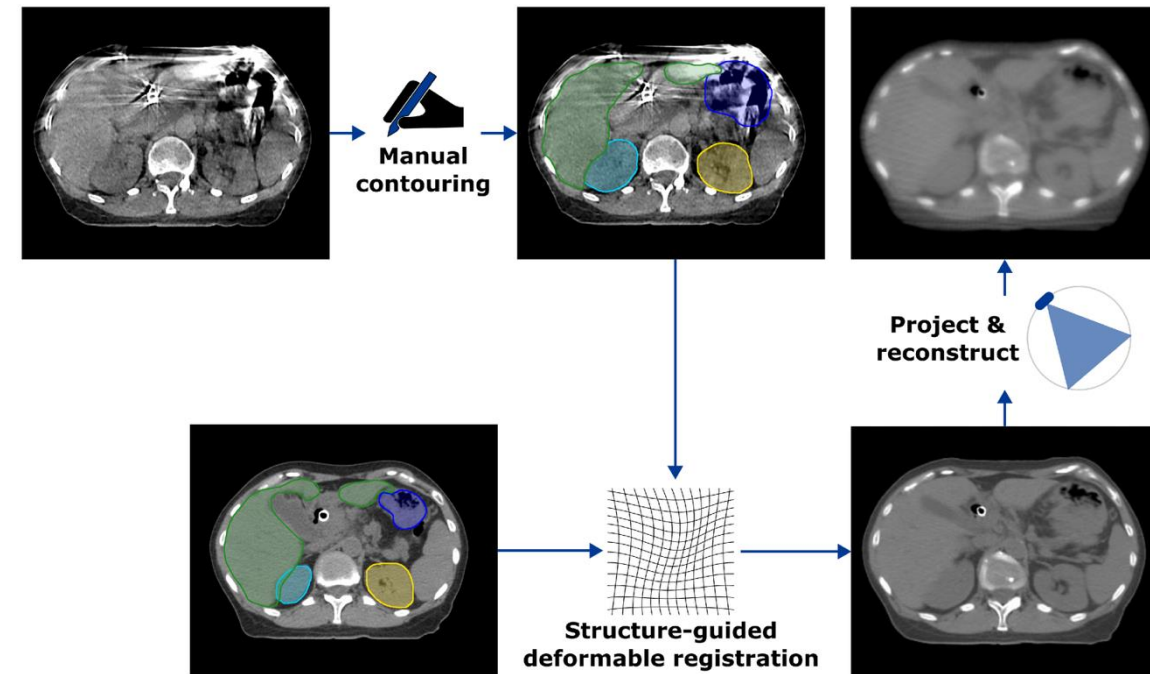
- MR-linac → Reference CT deformed to daily MRI
- Adaptive CBCT platform → Reference CT deformed to daily CBCT

CBCT-based sCT: e.g. Ethos



CBCT-based sCT: e.g. Ethos (non-Hypersight)

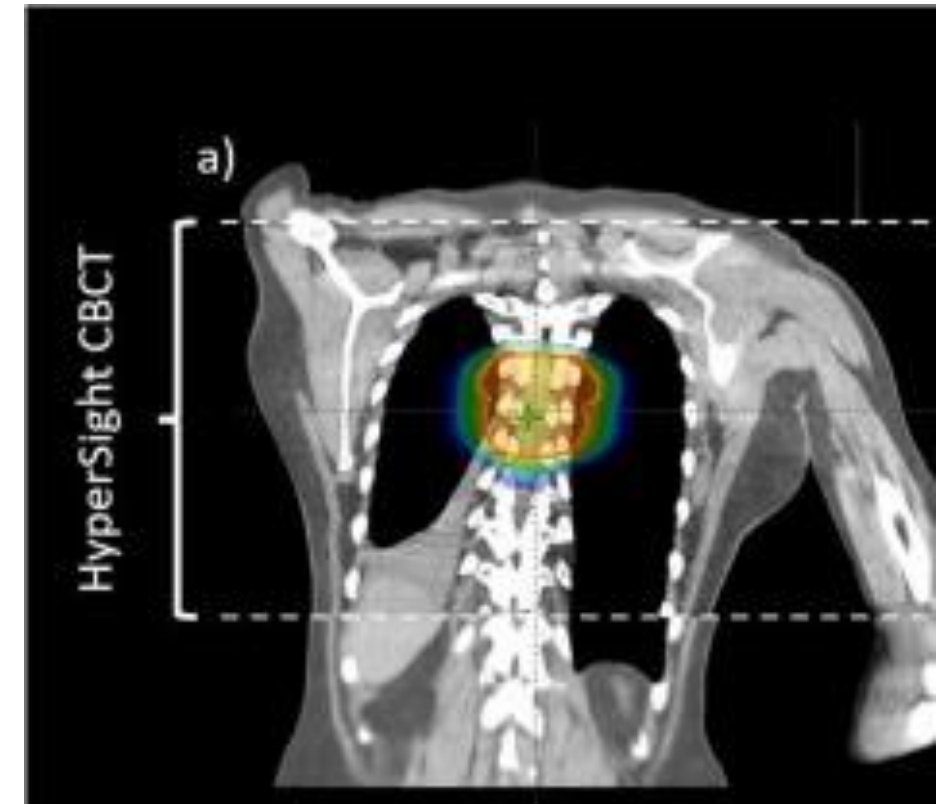
- Synthetic CT generation:
 - Planning CT deformed to CBCT
 - Technique = structure-guided DIR
 - Using DIR vector field, CBCT HUs replaced with planning CT HUs
 - Image volume derived from CBCT
 - HU values assigned from reference CT
- Planning CT is propagated outside CBCT FOV using rigid registration to compensate for limited FOV



Olberg et al. 2025

CBCT-based sCT: e.g. Ethos (Hypersight)

- Inside Hypersight CBCT FOV:
 - Direct dose calculation available.
- Outside Hypersight CBCT FOV:
 - Planning CT propagated outside CBCT FOV using rigid registration to compensate for limited FOV.
 - Note: stitching happens sup/inf but not laterally.



Hypersight vs. sCT

- 47 pelvis and breast plans analyzed
 - Close agreement between HS-iCBCT and sCT
 - Gamma passing rates $> 95\%$ (1%/1mm)
 - Some discrepancies in high gradient regions
- Trade-offs: potential dose differences
 - HS-iCBCT $\rightarrow$ imaging artifacts and FOV limitations
 - sCT $\rightarrow$ DIR limitations

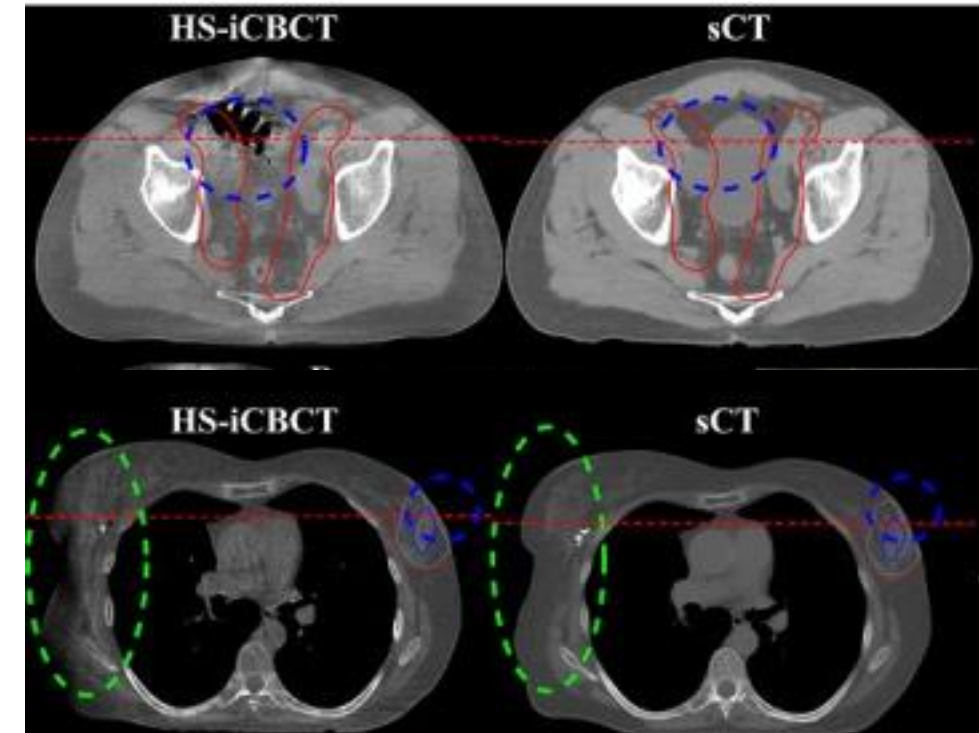


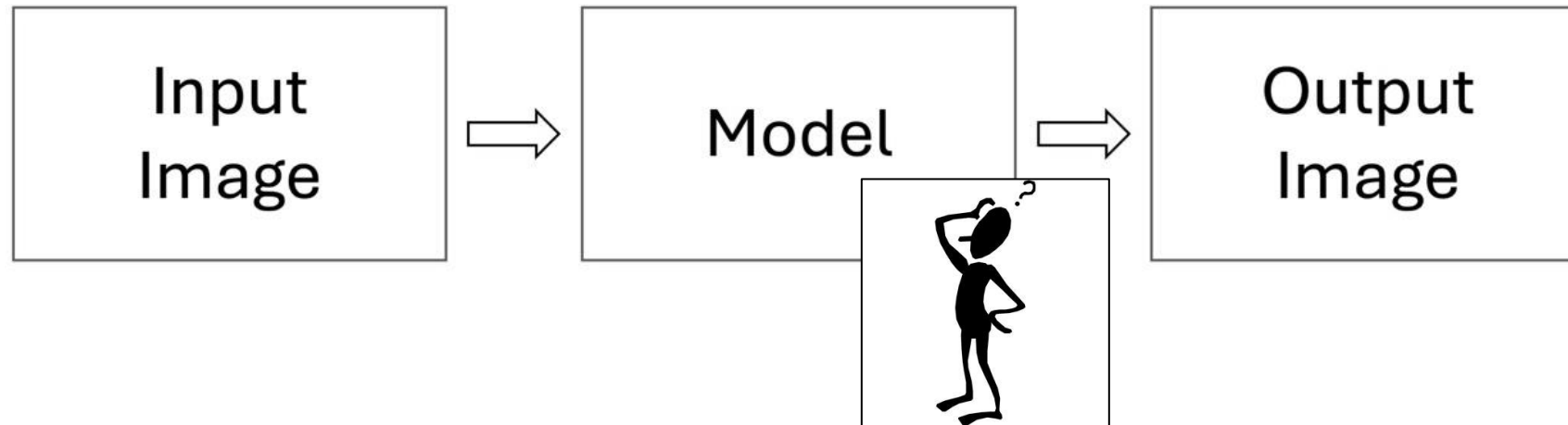
Table 5.2 Examples of commercial solutions available for synthetic CT reconstruction in offline and online adaptive radiotherapy.

Company	Product	Technique	MR- or CBCT-based sCT
Offline Adaptive			
GE Healthcare	MRI Planner (formerly Spectronics)	Deep learning	MR
Phillips	MRCAT	Bulk density, Deep learning	MR
RaySearch	RayStation	Hybrid (HU assignment and DIR)	CBCT
Siemens	Syngo.via	Bulk density, Deep learning	MR
Varian	Velocity	Deformable image registration	CBCT
Online Adaptive			
Elekta	Unity	Bulk density	MR
Varian	Ethos	Deformable image registration	CBCT
ViewRay Inc.	MRIdian	Hybrid (atlas- and voxel-based)	MR

Validation and QA

- Challenges
 - Generalizability of models
 - Lack of ground truth in absence of reference CT
- General approach
 1. **Model validation**
 2. **Patient-specific QA**
 3. **Machine-specific QA**

Model Validation



- For clinical user, model and training data are **under the hood**.
- Must validate that model produces accurate output images given clinic's input images.

Model Validation

- ICRP recommends total uncertainty in RT workflows within 5%.
Dose to PTV within 2% between CT- and sCT-based plans as an acceptance criterion
(Korsholm et al. 2014).
- 2% dose deviation threshold has been consistently adopted in practice across multiple sCT-based studies
(Dowling et al. 2012; Korhonen et al. 2014; Lambert et al. 2011).

Model Validation

- ✓ Compare real versus synthetic images in patient cohort.
- ✓ Validate that synthetic images correctly represent daily changes in anatomy.
- ✓ Determine patient eligibility criteria.

Model Validation

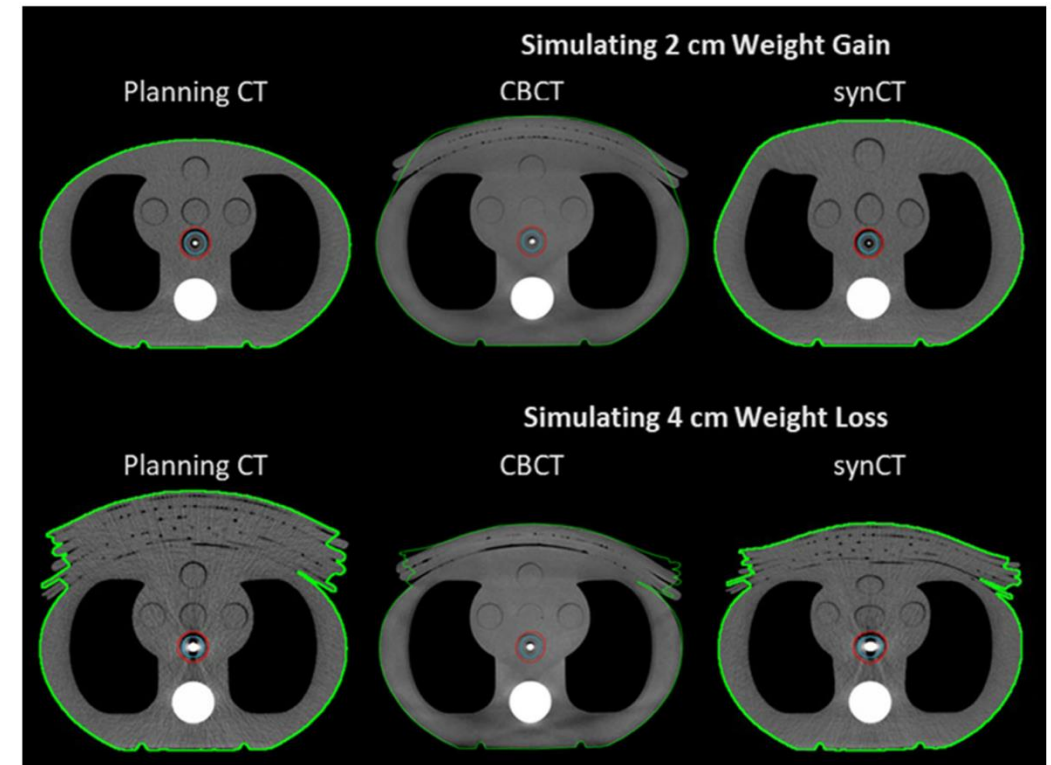
- ✓ Compare real versus synthetic images in cohort of patients.
 - ✓ Dose volume histograms, gamma metric, etc.
 - ✓ MR-OPERA Trial (Persson et al. 2017) and MR-PROTECT Trial (Persson et al. 2020):
 - ✓ Large-scale, multi-institutional validation of commercial sCT for prostate RT.
 - ✓ PTV mean dose differences within 1% considered clinically acceptable.
 - ✓ Deviations $>1\%$ were further investigated.
 - ✓ Minimum Gamma passing rates of $\geq 98\%$ (3%/2mm)

Model Validation (Adaptive)

- ✓ Validate sCT correctly represents daily changes in anatomy.
- ✓ Heterogeneous QA phantom modified during CBCT acquisition
- ✓ Simulate common clinical scenarios (weight change, target displacement, changes in gastrointestinal gas, etc.)
- ✓ sCTs robust to most clinical scenarios (ion chamber and gamma measurements).

Clinical commissioning of an adaptive radiotherapy platform: Results and recommendations

Kelly Kisling | Timothy D. Keiper | Daniela Branco | Grace Gwe-Ya Kim | Kevin L Moore | Xenia Ray

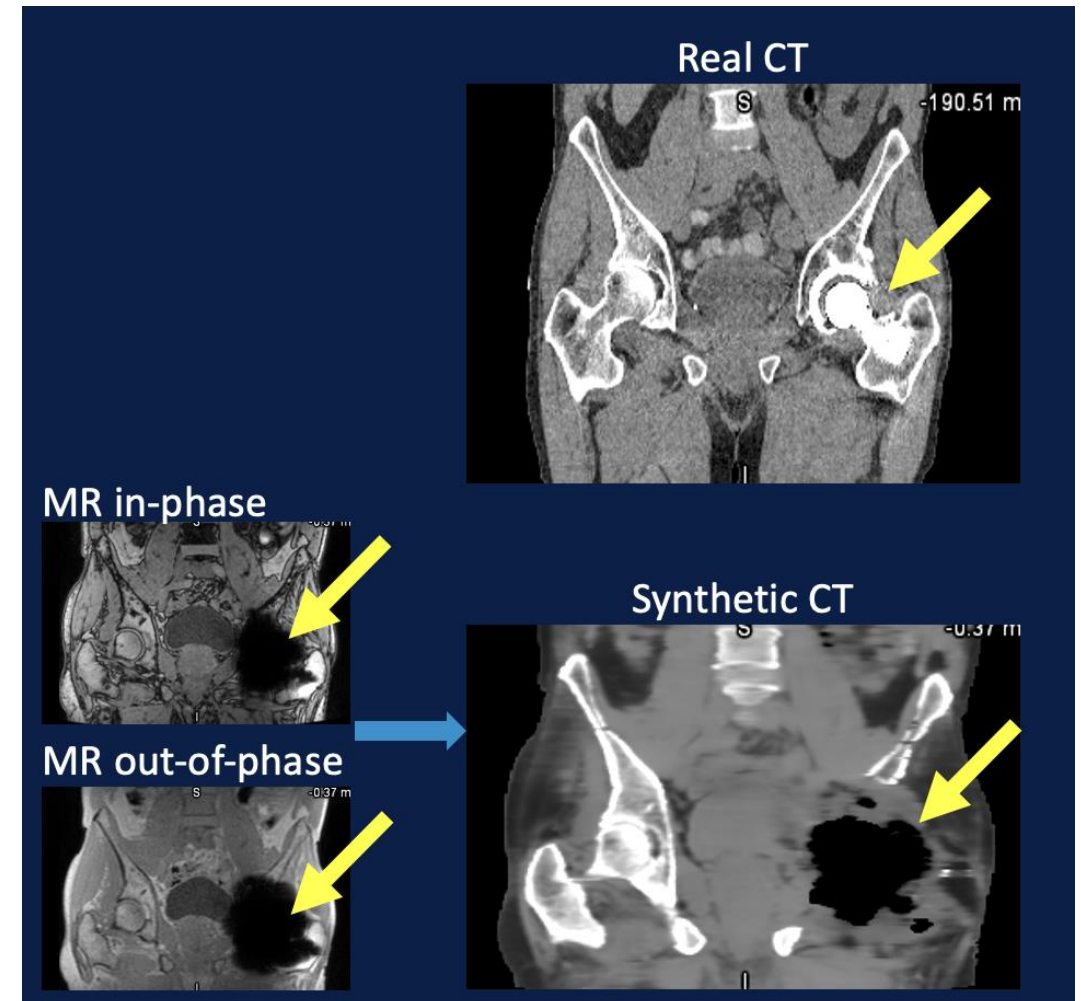


Model Validation

- ✓ Determine patient eligibility criteria

Examples:

- ✓ Implants and prostheses
- ✓ Weight limits



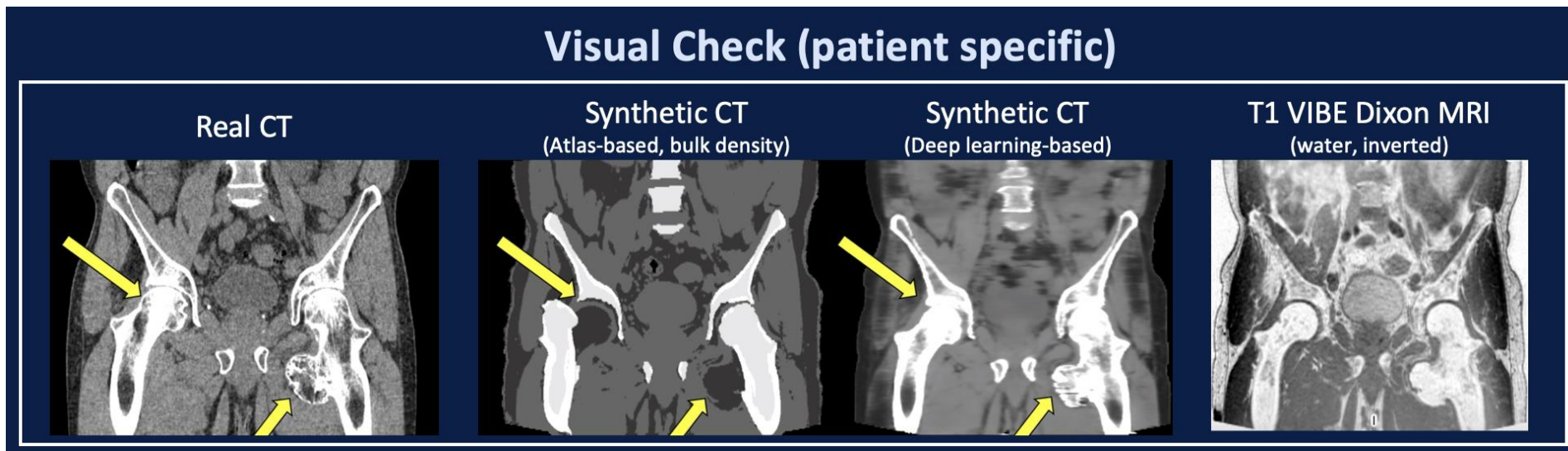
Model Validation

- Typical Validation workflow:
 - Paired (CT and sCT) datasets acquired in cohort of patients.
 - Forward-calculate clinical treatment plans onto both datasets.
 - Compare dose distributions.

- Summary of metrics:
 - Accept $\leq 2\%$ dose deviation between sCT and CT plans.
 - Enforce gamma passing rates $\geq 98\%$ (e.g. 3%/2mm, 15% cutoff).
 - For offline adaptive sCT, verify fiducial/CBCT setup accuracy.
 - Determine patient exclusion criteria.

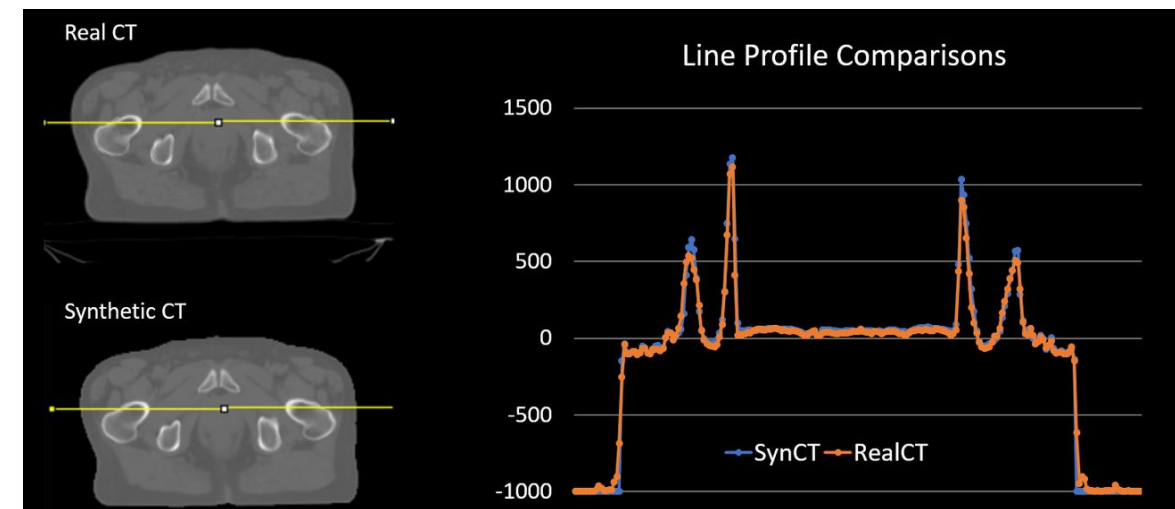
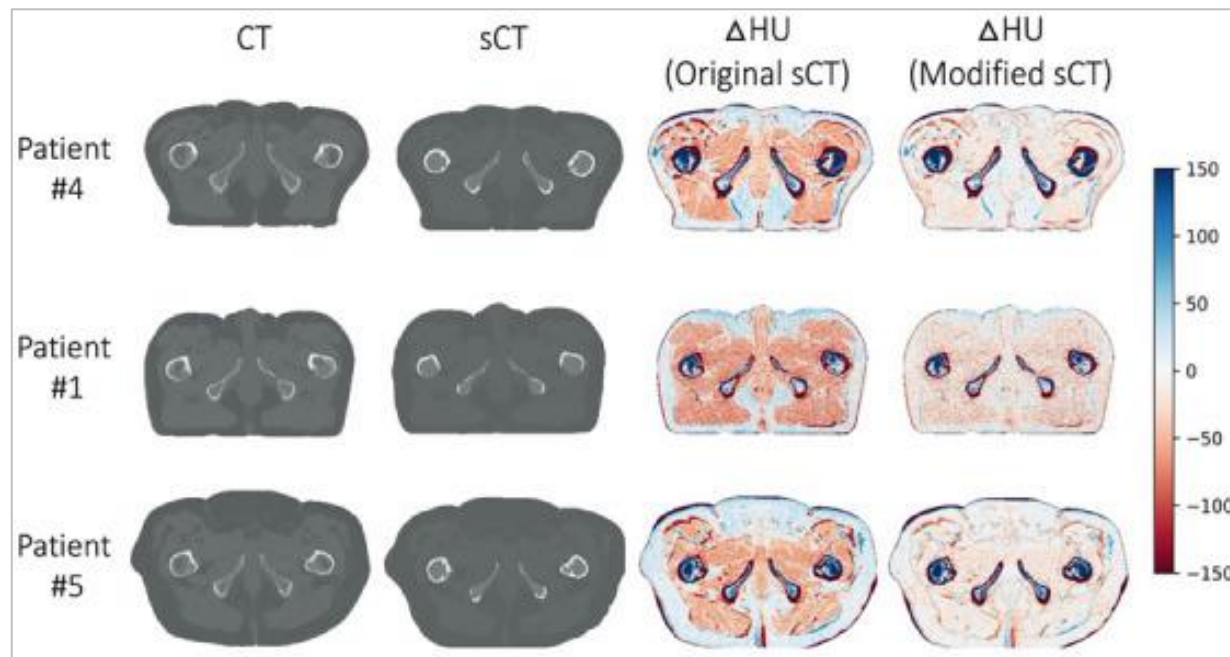
2. Patient specific QA

- Qualitative
 - CBCT-based and MRI-based sCT clinical users report visual assessment of **external body** and **high-density (e.g., bone) contours**.
 - 2cm threshold in external body contour reported (photon radiotherapy).
 - Ensure entire body contour included in FOV (if not, use care with beam selection).
- MRI: distortion correction and DICOM checks for correct acquisition.



2. Patient specific QA

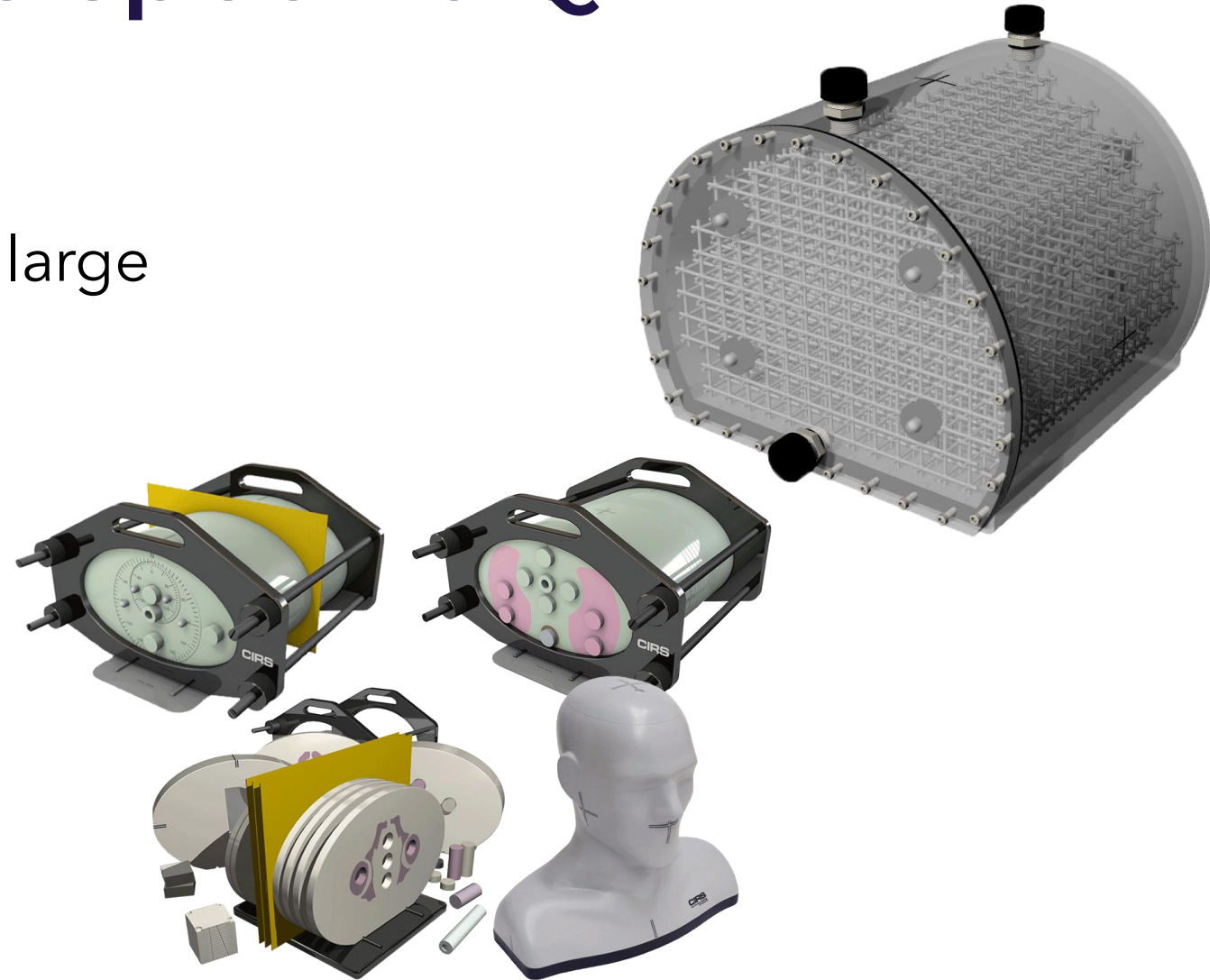
- Quantitative
 - Benchmark HU values using institutional data.
 - HU line profiles.



Centre Hospitalier de l'Université de Montréal, Canada

3. Machine specific QA

- Verify geometric fidelity using large field-of-view phantom.
 - Perform routinely (e.g. monthly)
- Test sCT reconstruction.
 - E.g. after system upgrades



Challenges and Future Directions

- Limited consensus guidelines on commissioning, validation, QA, etc.
- Future technical directions:
 - Faster / real-time synthetic imaging.
 - Increased generalizability and robustness to different scenarios (anatomical variability, scanner heterogeneity, etc.).
 - Large-scale, multi-institutional, and multivendor datasets.

Consensus Guidelines

Task Group No. 464: Recommendations for commissioning and quality assurance of synthetic computed tomography (sCT) for treatment planning and image guidance in photon radiotherapy (TG464)

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- Activities**
1. Review the motivation for using synthetic CT (sCT) clinically and existing methods for generating sCT data.
 2. Provide recommendations on the requirements of sCT data for clinical use in treatment planning and image guidance for photon treatment delivery.
 3. Provide recommendations on the commissioning, validation, and ongoing quality assurance (QA) of sCT.
 4. Provide an overview of potential challenges in incorporating sCT data in radiation therapy workflows.
 5. Provide recommendations on the risk-based approaches to quality management of sCT.

By-Laws: Not Referenced. **Rules:** [Not Referenced](#).

Standing? This is not a standing committee.

Approved Date(s): 2/15/2026 - 12/31/2028

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